

CYBERSECURITY INTERNSHIP REPORT

Week 3 – Advanced Security and Final Reporting

Submitted By

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1. Introduction

This report documents the activities performed during Week 3 of the Cybersecurity Internship Program. The objective of this task was to perform basic penetration testing, implement application logging, create a security checklist, and prepare final project documentation.

The work was carried out on a secure Node.js web application developed during Week 2.

2. Objectives

The objectives of this task were:

- Perform basic penetration testing.
- Learn basic network reconnaissance techniques.
- Implement application logging using Winston.
- Create a web application security checklist.
- Document findings and improvements.
- Prepare project deliverables for final submission.

3. Testing Environment

Component	Description
Operating System	Kali Linux
Application	Secure Authentication Demo
Runtime Environment	Node.js
Framework	Express.js
Testing Tool	Nmap
Logging Tool	Winston
API Testing Tool	Postman

4. Basic Penetration Testing

4.1 Nmap Scan

A basic network scan was performed using Nmap to identify open ports and active services.

Command Used

```
nmap localhost
```

Result

The scan successfully detected an open TCP port.

Output

```
PORT      STATE SERVICE  
3000/tcp  open  ppp
```

Observation

Port 3000 was identified as open, indicating that the Express.js web application was actively listening for incoming connections.

Purpose

The scan verified service availability and identified exposed network services as part of a basic security assessment.

4.2 Browser-Based Security Testing

Basic browser testing was performed to verify application behavior.

Tests Conducted

- Invalid email submission
- Invalid password submission
- Authentication testing

- JWT token verification

Result

The application correctly validated user input and handled authentication requests securely.

Screenshot

[Insert Browser Testing Screenshot Here]

5. Application Logging Using Winston

Description

Winston logging was implemented to monitor application activities and authentication events.

Installation

```
npm install winston
```

Configuration

```
const winston = require('winston');

const logger = winston.createLogger({
  transports: [
    new winston.transports.Console(),
    new winston.transports.File({
      filename: 'security.log'
    })
  ]
});
```

Logging Events

The application records:

- Application startup
- Login attempts
- Successful logins
- Failed login attempts

Example Log Entries

Application started

Login attempt from test@gmail.com

Successful login for test@gmail.com

Login attempt from attacker@gmail.com

Failed login attempt for attacker@gmail.com

Benefits

- Activity monitoring
- Security auditing
- Incident investigation
- Authentication tracking

6. Security Checklist

A security checklist was created to document security best practices for web application development.

Security Checklist

Input Validation

- Validate all user inputs.
- Reject malformed email addresses.
- Enforce password requirements.

Authentication

- Use JWT authentication.
- Protect secret keys.
- Implement token expiration.

Password Security

- Hash passwords using bcrypt.
- Never store passwords in plain text.
- Use salting techniques.

Secure Communication

- Use HTTPS for data transmission.
- Protect sensitive information.

Security Headers

- Use Helmet.js.
- Enable Content Security Policy (CSP).
- Prevent clickjacking attacks.

Logging and Monitoring

- Log authentication events.
- Monitor suspicious activities.
- Review security logs regularly.

General Security

- Keep dependencies updated.
- Remove debugging information.
- Perform regular security testing.

7. Results

Activity	Status
Nmap Scan	Completed
Browser Testing	Completed
Winston Logging	Implemented
Security Checklist	Created
Documentation	Completed

8. Key Achievements

The following improvements were achieved during Week 3:

- Performed basic penetration testing.
- Verified application network exposure.
- Implemented centralized logging.
- Recorded authentication events.
- Created a security best-practices checklist.
- Improved monitoring and auditing capabilities.

9. Conclusion

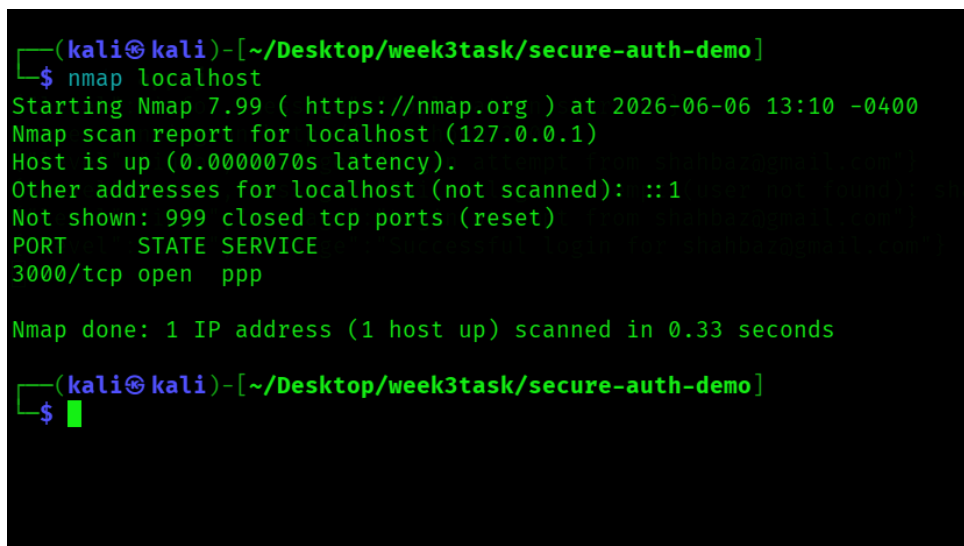
Week 3 focused on strengthening security monitoring and documenting security practices.

Basic penetration testing using Nmap confirmed that the application was running correctly and exposed only the intended service. Winston logging was successfully implemented to record important application events and authentication activities. A comprehensive security checklist was also created to promote secure development practices.

These activities enhanced understanding of security monitoring, basic penetration testing, and web application security management.

Screenshots

Screenshot 1 – Nmap Scan Result

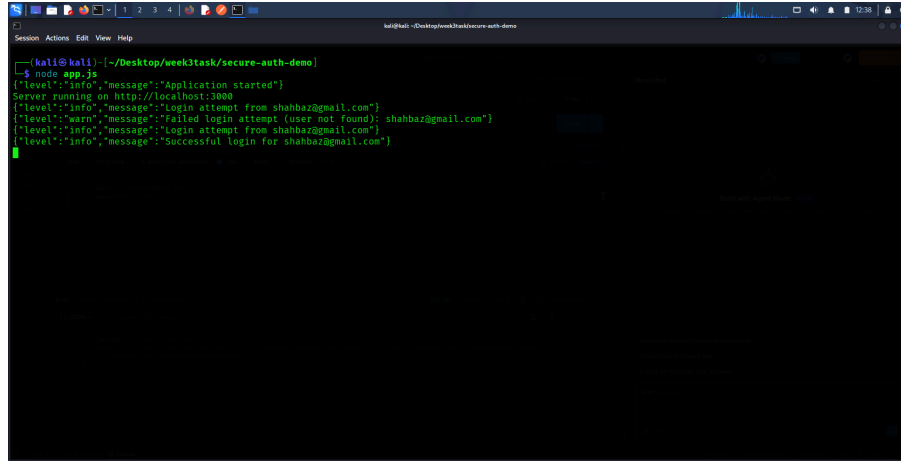


```
(kali㉿kali)-[~/Desktop/week3task/secure-auth-demo]
└─$ nmap localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-06 13:10 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000070s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
3000/tcp  open  ppp
3001/tcp  open  ppp

Nmap done: 1 IP address (1 host up) scanned in 0.33 seconds

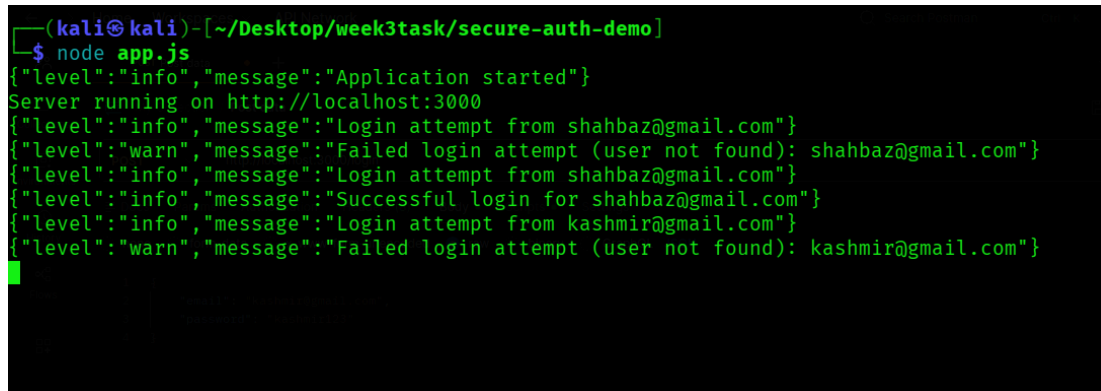
(kali㉿kali)-[~/Desktop/week3task/secure-auth-demo]
└─$
```

Screenshot 2 – Application Running

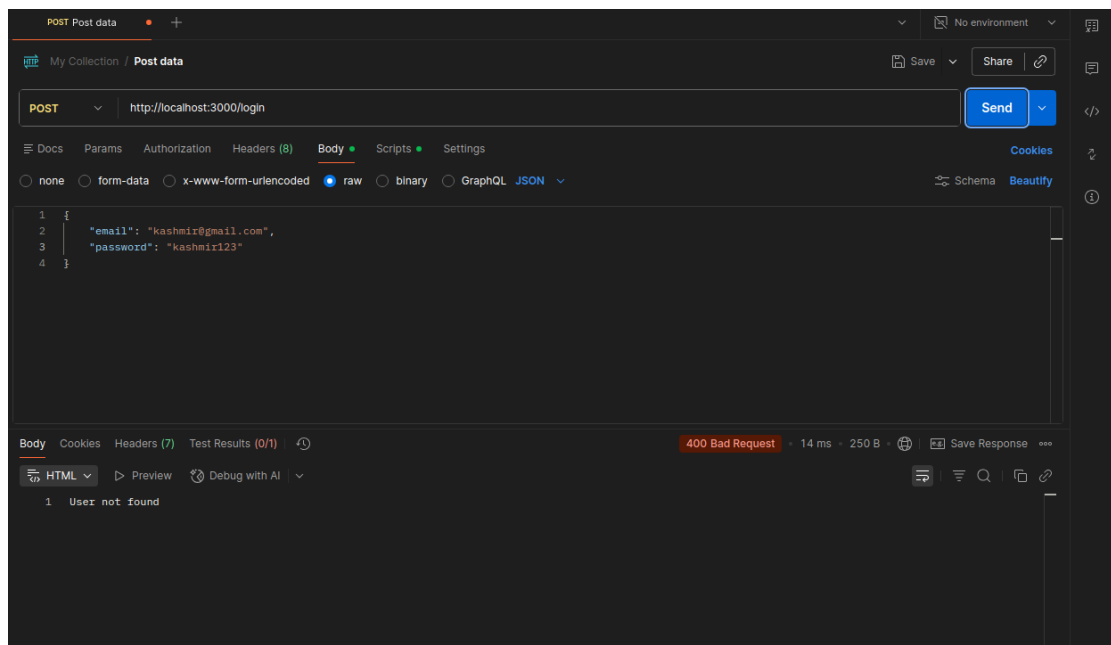


```
(kali@kali) [~/Desktop/week3task/secure-auth-demo]
$ node app.js
{"level":"info","message":"Application started"}
Server running on http://localhost:3000
{"level":"info","message":"Login attempt from shahbaz@gmail.com"}
{"level":"warn","message":"Failed login attempt (user not found): shahbaz@gmail.com"}
{"level":"info","message":"Login attempt from shahbaz@gmail.com"}
{"level":"info","message":"Successful login for shahbaz@gmail.com"}
{"level":"info","message":"Login attempt from shahbaz@gmail.com"}
{"level":"warn","message":"Failed login attempt (user not found): shahbaz@gmail.com"}
```

Screenshot 3 – Invalid Email



```
(kali@kali) [~/Desktop/week3task/secure-auth-demo]
$ node app.js
{"level":"info","message":"Application started"}
Server running on http://localhost:3000
{"level":"info","message":"Login attempt from shahbaz@gmail.com"}
{"level":"warn","message":"Failed login attempt (user not found): shahbaz@gmail.com"}
{"level":"info","message":"Login attempt from shahbaz@gmail.com"}
{"level":"info","message":"Successful login for shahbaz@gmail.com"}
{"level":"info","message":"Login attempt from kashmir@gmail.com"}
{"level":"warn","message":"Failed login attempt (user not found): kashmir@gmail.com"}
```



Screenshot 4 – Valid Email

```
(kali㉿kali)-[~/Desktop/week3task/secure-auth-demo]
└─$ node app.js
{"level":"info","message":"Application started"}
Server running on http://localhost:3000
{"level":"info","message":"Login attempt from shahbaz@gmail.com"}
{"level":"warn","message":"Failed login attempt (user not found): shahbaz@gmail.com"}
{"level":"info","message":"Login attempt from shahbaz@gmail.com"}
{"level":"info","message":"Successful login for shahbaz@gmail.com"}
└─$
```

Screenshot 5 – Security Checklist

```
(kali㉿kali)-[~/Desktop/week3task/secure-auth-demo]
└─$ ls
app.js  node_modules  package.json  package-lock.json  security.log

(kali㉿kali)-[~/Desktop/week3task/secure-auth-demo]
└─$ cat security.log
{"level":"info","message":"Application started"}
{"level":"info","message":"Application started"}
{"level":"info","message":"Application started"}
{"level":"info","message":"Application started"}
{"level":"info","message":"Login attempt from shahbaz@gmail.com"}
{"level":"warn","message":"Failed login attempt (user not found): shahbaz@gmail.com"}
{"level":"info","message":"Login attempt from shahbaz@gmail.com"}
{"level":"warn","message":"Failed login attempt (user not found): shahbaz@gmail.com"}
{"level":"info","message":"Application started"}
{"level":"info","message":"Application started"}
{"level":"info","message":"Application started"}
{"level":"info","message":"Application started"}
{"level":"info","message":"Login attempt from shahbaz@gmail.com"}
{"level":"warn","message":"Failed login attempt (user not found): shahbaz@gmail.com"}
{"level":"info","message":"Login attempt from shahbaz@gmail.com"}
{"level":"info","message":"Successful login for shahbaz@gmail.com"}
└─$
```